

CPE 445-Internet of Things

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About the Instructor

- Post Doc, LPWAN, 2017
 - Glasgow Caledonian University, UK.
- PhD, IoT, 2016
 - Glasgow Caledonian University, UK.
- MSc Control Systems, 2008
 - University of Sheffield, UK.
- BSc, Computer Engineering, 2006
 - COMSATS IIT, Lahore, Pakistan.

Motivation:

Company: Punjab Information Technology Board, Govt. of Punjab Website: <https://pitb.gov.pk/>

Typical Day at Work:

- Identifying potential problems in government and private sectors.
- Suggesting a viable solution based on latest tech implementation in embedded, electronics and circuit design (Arduino, Raspberry Pi, Adafruit, TI, Hard kernel Odroid).
- Integration of different sensor and modules with controllers.
- Implementation of pre-trained AI/ML Model, preferably in Computer Vision.
- Debugging and testing of electronic circuits and mechanical systems
- Continuous improvement of deployed systems (turnstiles, RFID Access System etc.)

You are required to have:

- Understanding Datasheets and Specs details.
- Knowledge of communication protocols in embedded system (I2C, SPI, UART, Wiegand, USB)
- Knowledge of working principle of RF communication (LoRa, NFC, RFID, BLE, Nb-IoT)
- Preparing prototypes for pilot project as a proof of concept. • Grasp on Data structure, algorithm and basics of C++/C#/Python.
- Hands-on experience of implementing and configuring hardware based solution (CCTV, Access Control, Parking)

Contents

- Genesis of IoT: This section highlights IoT's place in the evolution and development of the Internet.
- IoT and Digitization: This section details the differences between IoT and digitization and defines a framework for better understanding their relationship.
- IoT Impact: This section shares a few high-level scenarios and examples to demonstrate the influence IoT will have on our world.
- Convergence of IT and OT: This section explores how IoT is bringing together information technology (IT) and operational technology (OT).
- IoT Challenges: This section provides a brief overview of the difficulties involved in transitioning to an IoT-enabled world.

What is IoT?

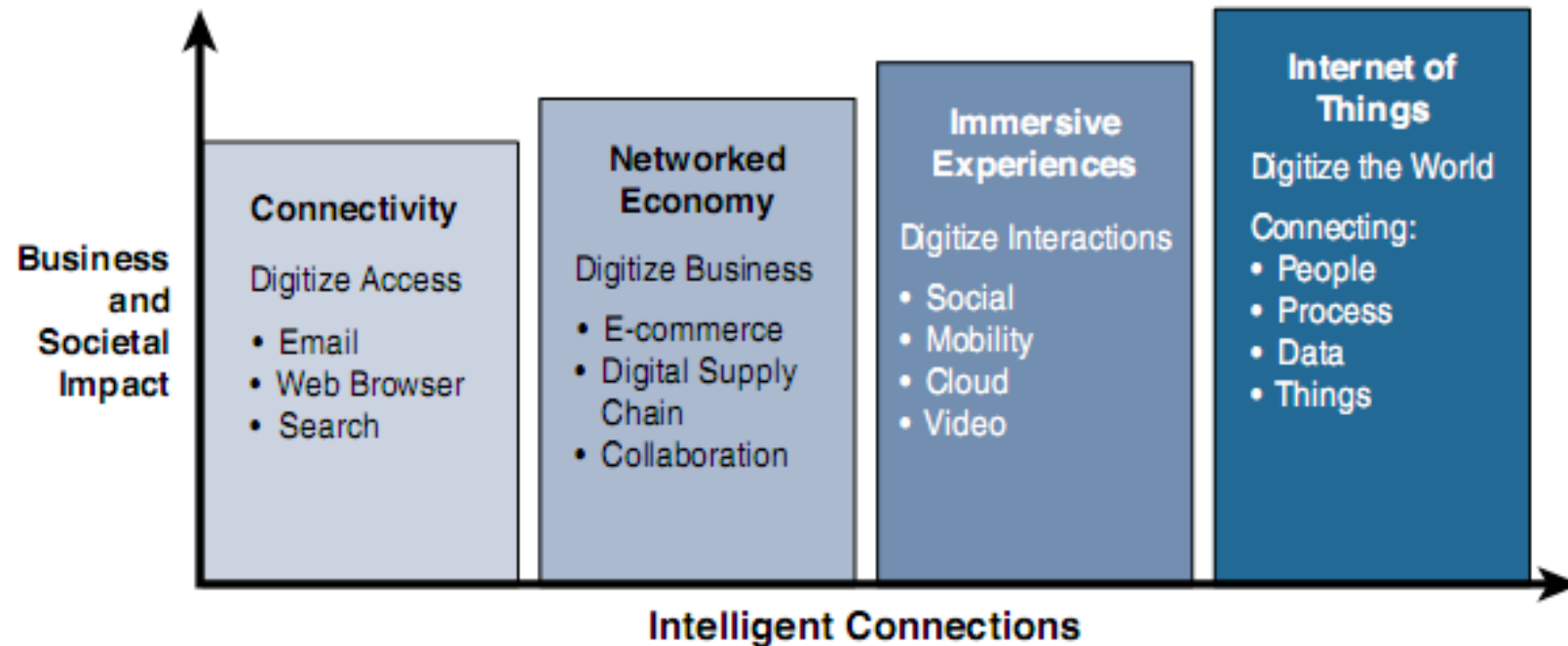


A phenomenon which connects a variety of things
– Everything that has the ability to communicate

Genesis of IoT

- Kevin Ashton created the term “Internet of Things” while working for Proctor & Gamble in 1999
- Age of IoT started between 2008 and 2009
- The basic premise and goal of IoT is to “connect the unconnected”
- IoT is a technology transition in which devices will allow us to sense and control the physical world by making objects smarter and connecting them through an intelligent network
- IoT is broad and multifaceted

Evolution of IoT



Evolutionary Phases of the Internet

- **Connectivity (Digitize Access)**-This phase connected people to email, web services, and search so that information is easily accessed
- **Networked Economy (Digitize business)**- This phase enabled e-commerce and supply chain enhancements along with collaborative engagement to drive increased efficiency in business processes
- **Immersive Experiences (Digitize interactions)**- This phase extended the Internet experience to encompass widespread video and social media while always being connected through mobility. More and more applications are moved into the cloud
- **Internet of Things (Digitize the world)**- This phase is adding connectivity to objects and machines in the world around us to enable new services and experiences. It is connecting the unconnected

IoT and Digitization

- IoT and digitization are terms that are often used interchangeably
- IoT focuses on connecting “things,” such as objects and machines, to a computer network, such as the Internet
- Digitization can mean different things to different people but generally encompasses the connection of “things” with the data they generate and the business insights that result
 - Wifi Tracking
 - Photography, DVD Rental etc.
 - Transportation Industry- Uber, Careem etc.
 - Nest

Any--X Point of View

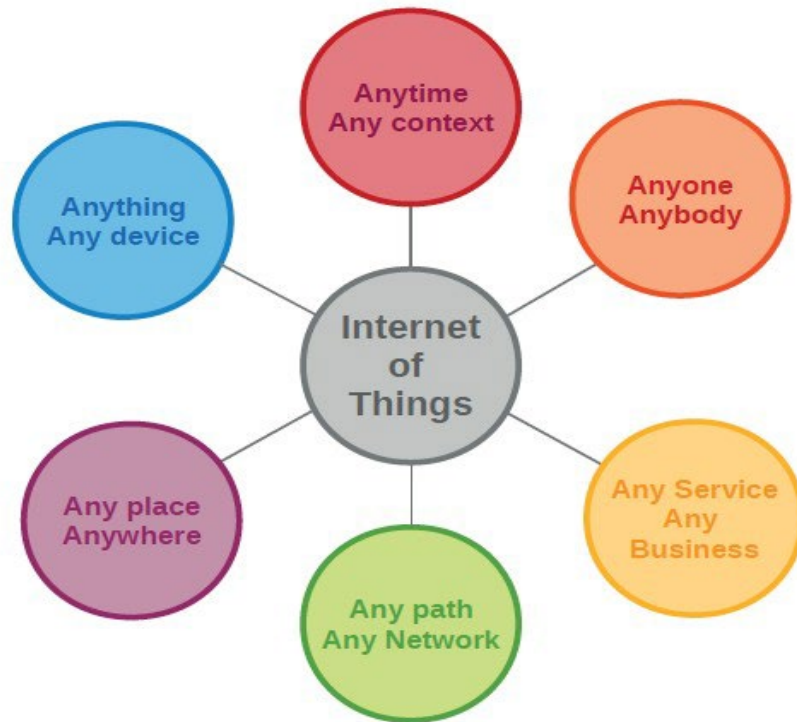
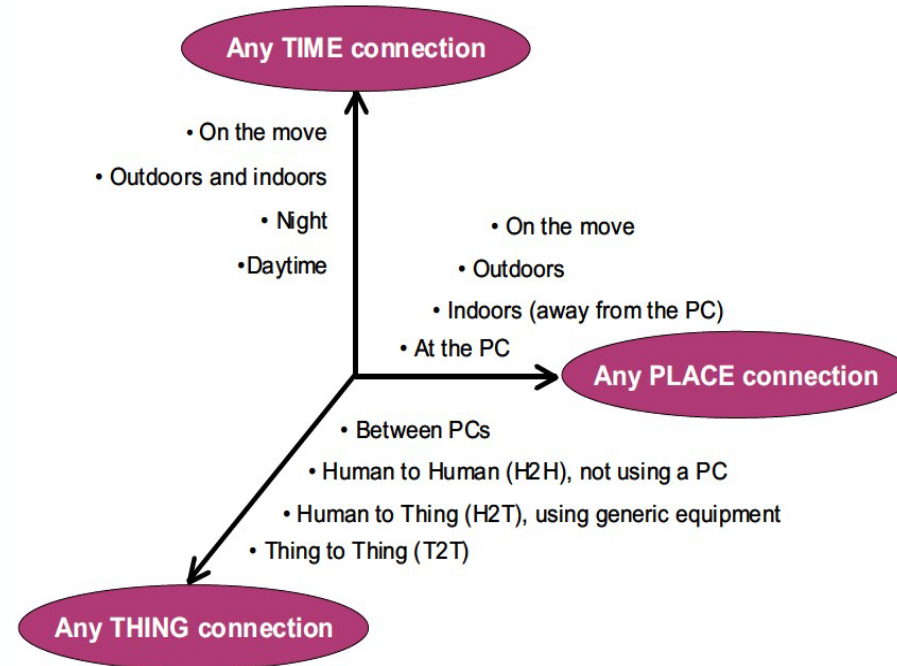


Figure 1 – A new dimension



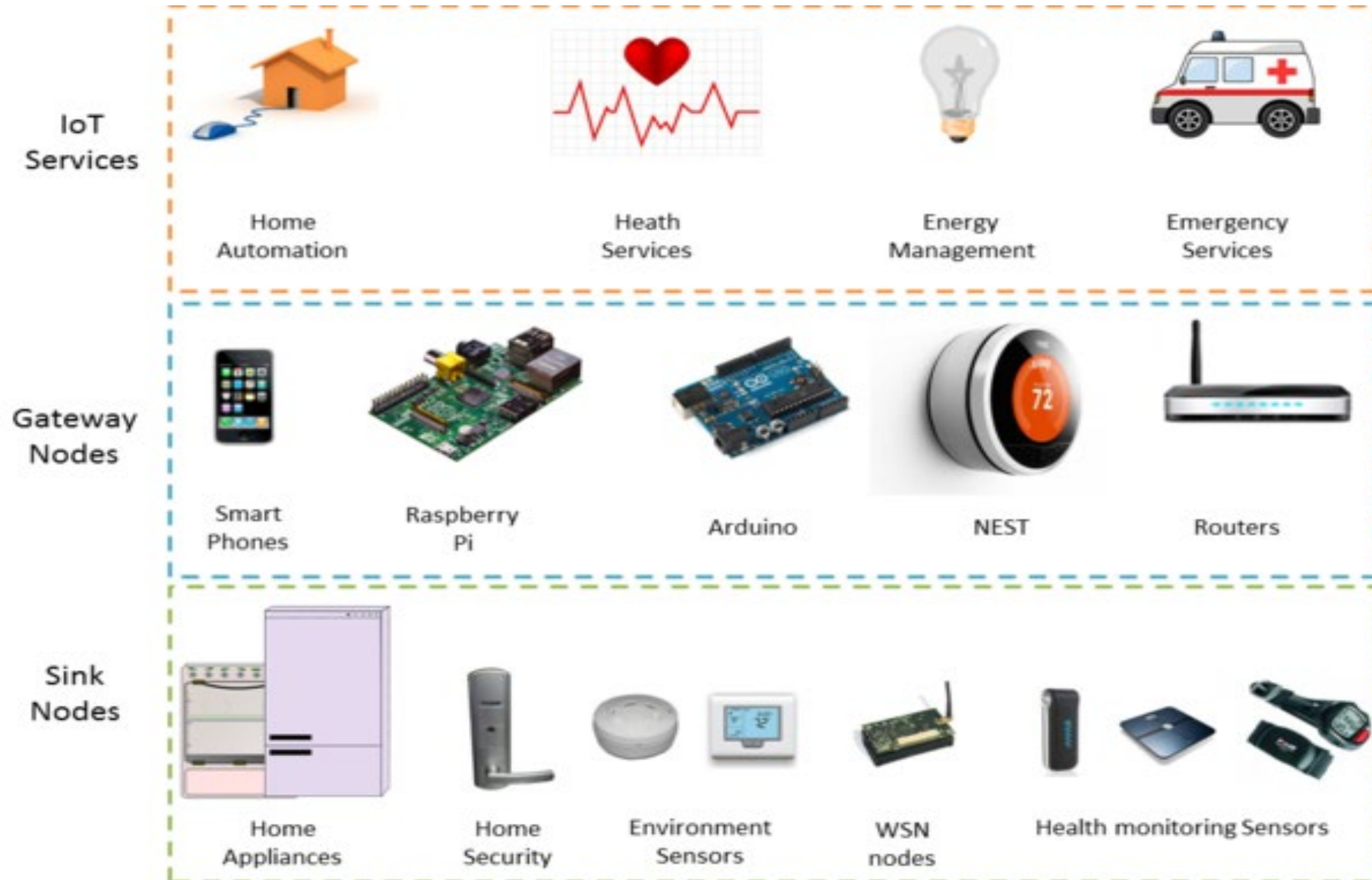
Source: ITU adapted from Nomura Research Institute

The Internet of Things allows people and things to be connected Anytime, Anyplace, with Anything and Anyone, ideally using Any path/ network and Any service.

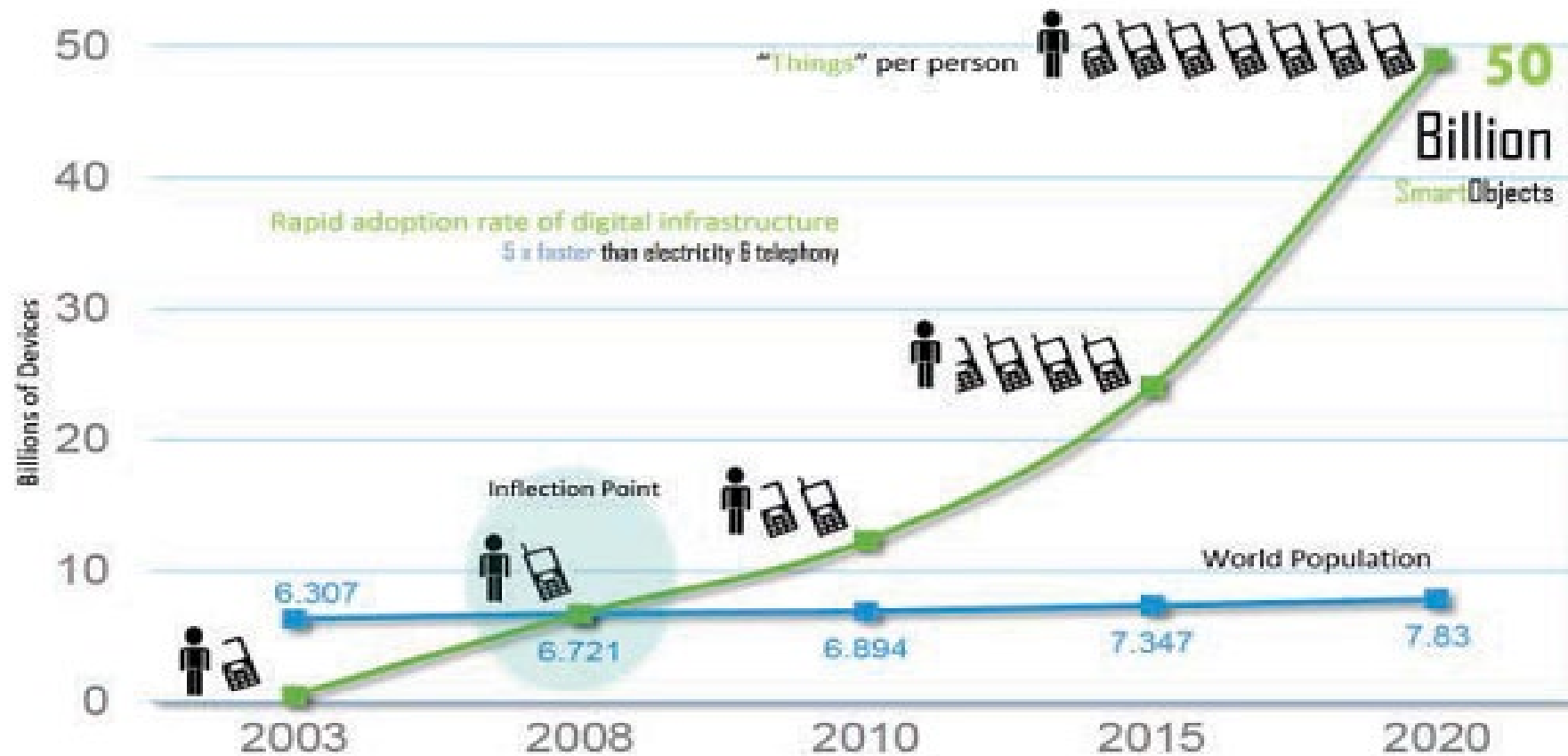
Characteristics of IoT

1. Intelligence
 - Knowledge extraction from the generated data
2. Architecture
 - A hybrid architecture supporting many others
3. Complex system
 - A diverse set of dynamically changing objects
4. Size considerations
 - Scalability
5. Time considerations
 - Billions of parallel and simultaneous events
6. Space considerations
 - Localization
7. Everything--as--a--service
 - Consuming resources as a service

Architecture for Open IoT Services



IoT Impact



The Rapid Growth in the Number of Devices Connected to the Internet

IoT Impact- Connected Roadways

- Self-driving vehicles- Google's self-driving vehicle
- Fully connected transportation infrastructure
- Self-driving vehicles need always-on, reliable communications and data from other transportation-related sensors to reach their full potential

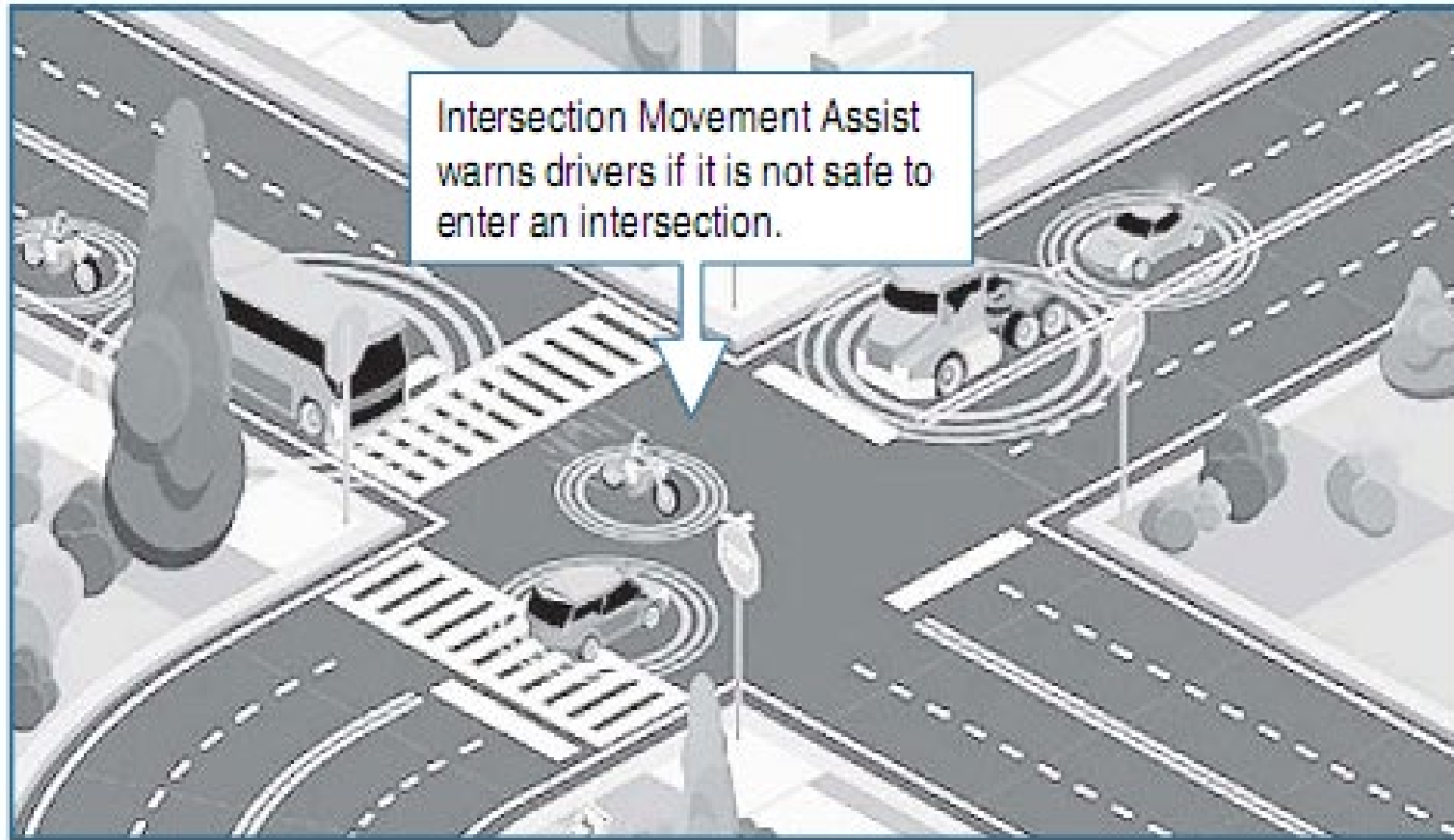


Google's Self-Driving Car

IoT Impact- Current Challenges Being Addressed by Connected Roadways

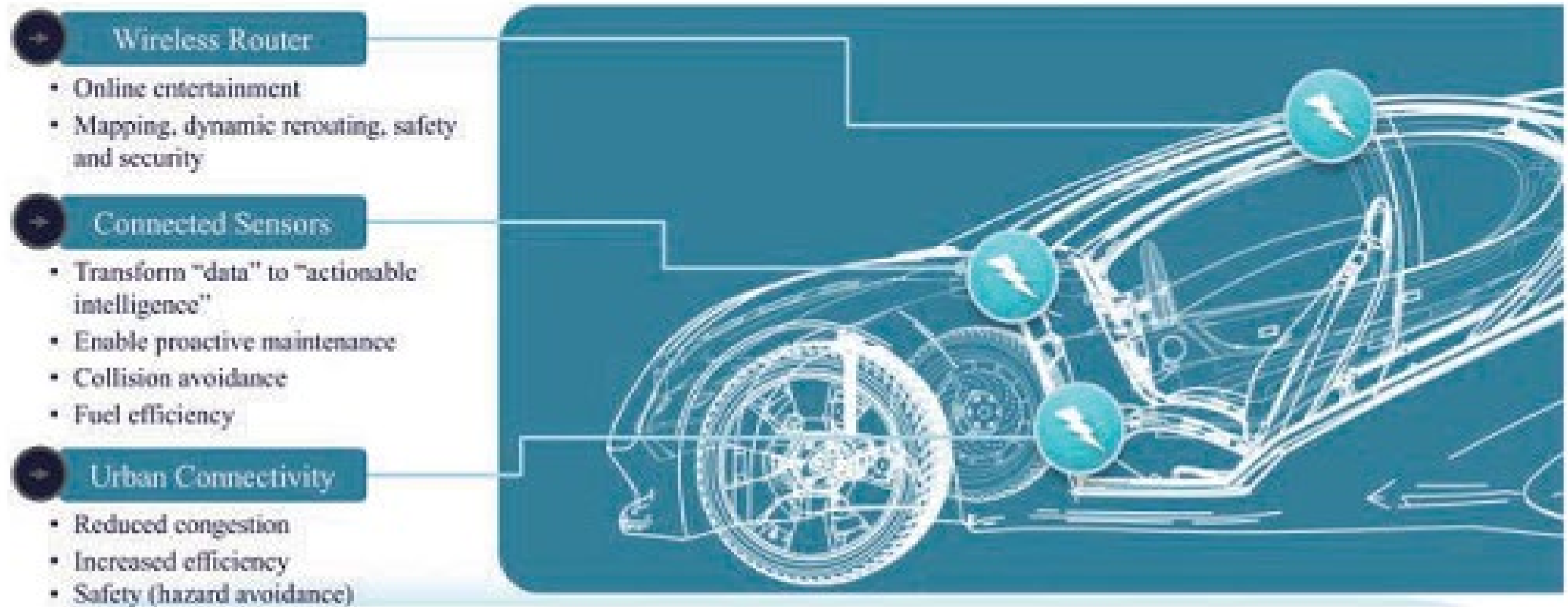
Challenge	Supporting Data
Safety	According to the US Department of Transportation, 5.6 million crashes were reported in 2012 alone, resulting in more than 33,000 fatalities. IoT and the enablement of connected vehicle technologies will empower drivers with the tools they need to anticipate potential crashes and significantly reduce the number of lives lost each year
Mobility	More than a billion cars are on the roads worldwide. Connected vehicle mobility applications can enable system operators and drivers to make more informed decisions, which can, in turn, reduce travel delays. Congestion causes 5.5 billion hours of travel delay per year, and reducing travel delays is more critical than ever before. In addition, communication between mass transit, emergency response vehicles, and traffic management infrastructures help optimize the routing of vehicles, further reducing potential delays
Environment	According to the American Public Transportation Association, each year transit systems can collectively reduce carbon dioxide (CO₂) emissions by 16.2 million metric tons by reducing private vehicle miles. Connected vehicle environmental applications will give all travelers the real-time information they need to make “green” transportation choices

IoT Impact- Connected Roadways



Application of Intersection Movement Assist

IoT Roadmap-Connected Car



The Connected Car

How much data will be generated by a connected car?

25 GB per hour

IoT Impact-Connected Roadways

- Few other applications of IoT for roadways are:
 - **Vehicle Tracking:** a vehicle's location is used for notification of arrival times, theft prevention, or highway assistance
 - **Cargo Management:** Cargo management provides precise positioning of cargo as it is en route so that notification alerts can be sent to a dispatcher and routes can be optimized for congestion and weather
 - **Road Weather Communication:** Road weather communications use sensors and data from satellites, roads, and bridges to warn vehicles of dangerous conditions or inclement weather on the current route

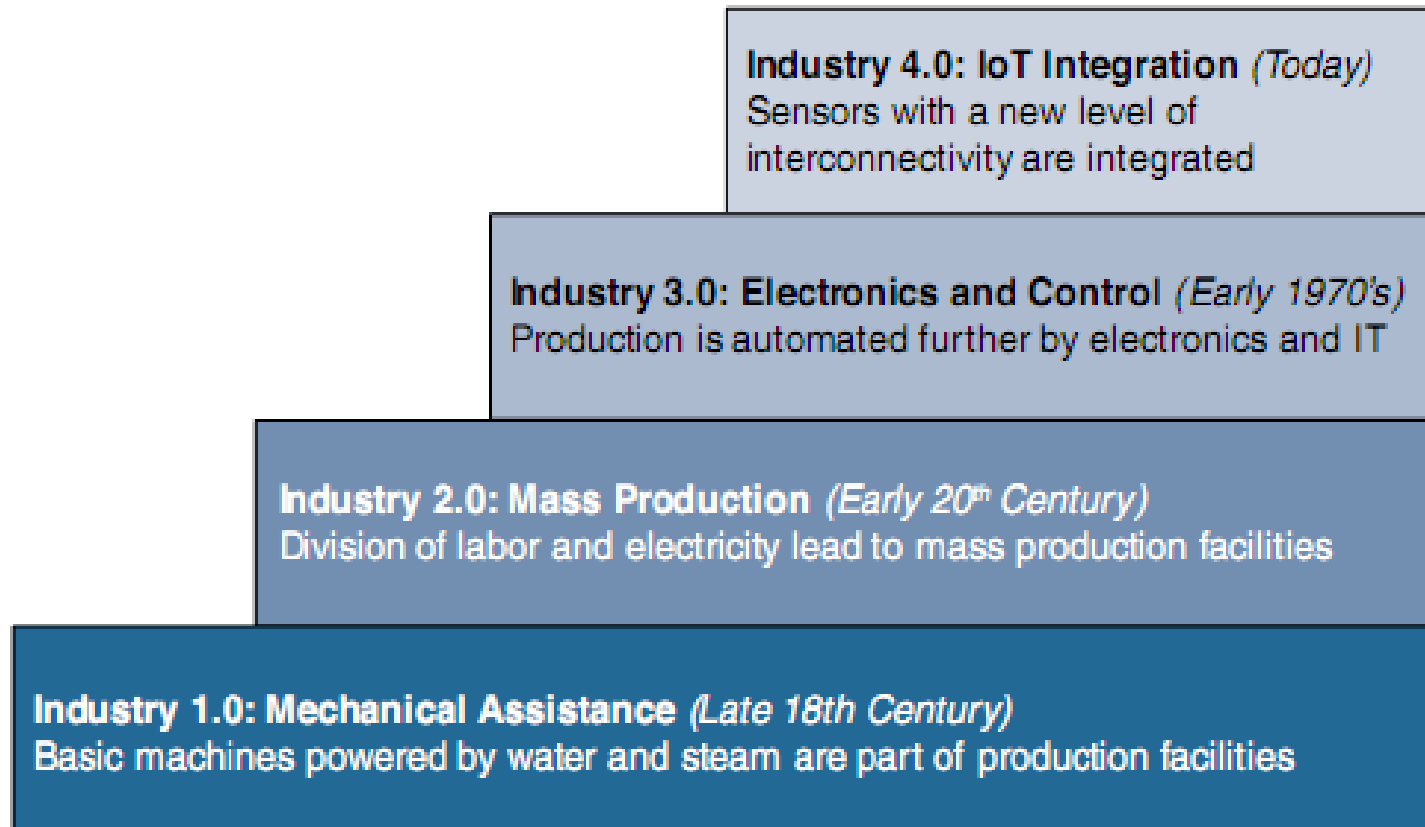
IoT Impact: Connected Factory

- Challenges facing manufacturing in a factor environment
 - Accelerating new product and service introductions to meet customer and market opportunities
 - Increasing plant production, quality, and uptime while decreasing cost
 - Mitigating unplanned downtime (which wastes, on average, at least 5% of production)
 - Securing factories from cyber threats
 - Decreasing high cabling and re-cabling costs (up to 60% of deployment costs)
 - Improving worker productivity and safety

IoT Impact: Connected Factory

- Convergence of factory-based operational technologies and architectures with global IT networks is referred to as the connected factory
- Large number of sensors are already used in factories. With IoT, these sensors are becoming advanced and attaining new level of connectivity (mainly using Internet Protocol (IP))
- Devices on plant floor transmit and receive large quantities of real-time informational and diagnostic data
- Connected Factory Examples:
 - “machine-to-people” connections are implemented to bring sensor data directly to operators on the floor via mobile devices
 - real-time location system (RTLS) utilizes small and easily deployed Wi-Fi RFID tags that attach to virtually any material and provide real-time location and status

IoT Impact: Industry Revolution

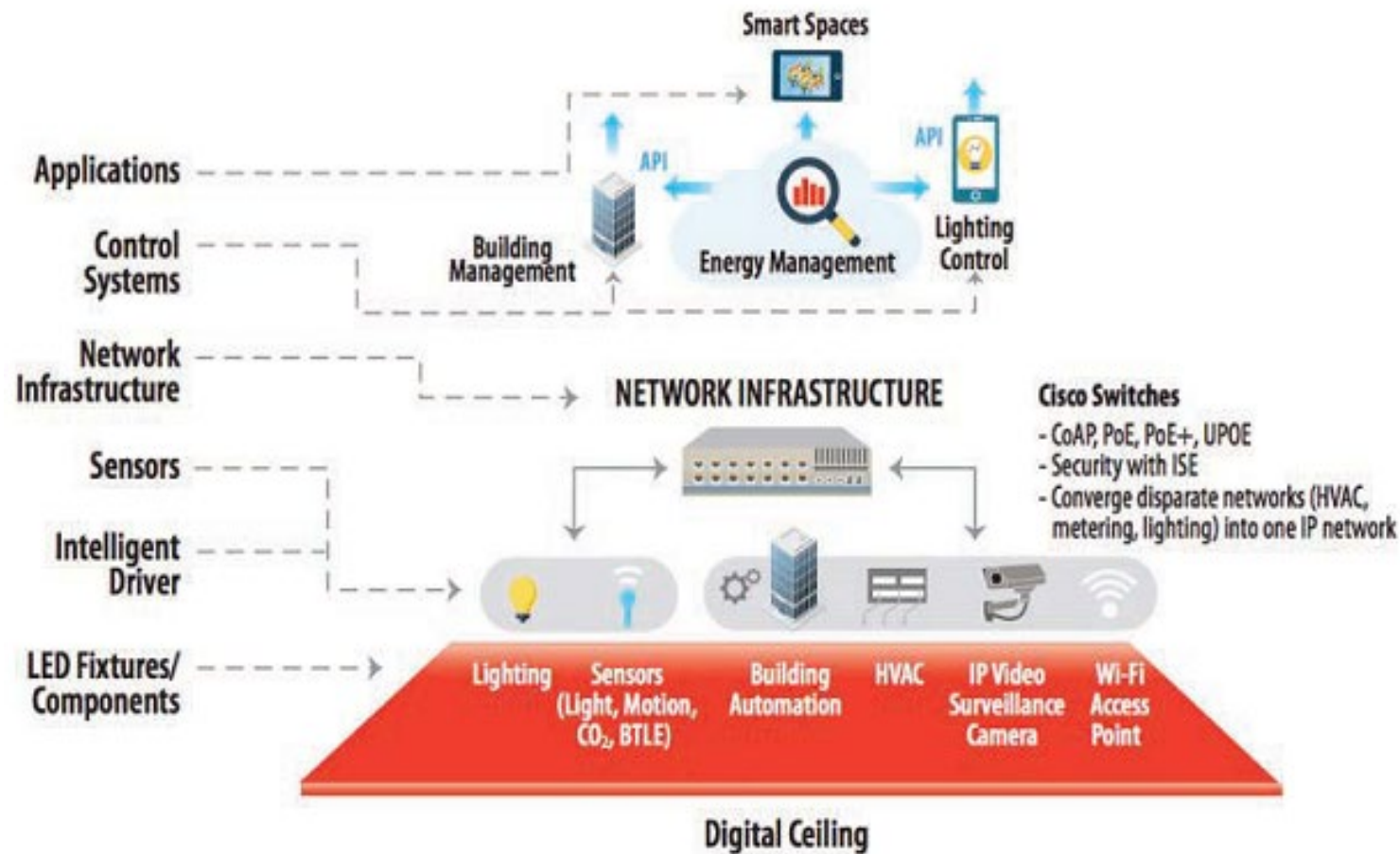


The Four Industrial Revolutions

Industry 4.0

- Industry 4.0 takes manufacturing from a purely automated assembly line model of production to a model where the machines are intelligent and communicate with one another
- People and machines work together to analyze the data and make intelligent decisions
- Machines predicting failures and self-healing
- Human monitoring and intervention are no longer necessary

IoT Impact- Smart Connected Buildings



A Framework for the Digital Ceiling

Smart Connected Building- LED Digital Ceiling

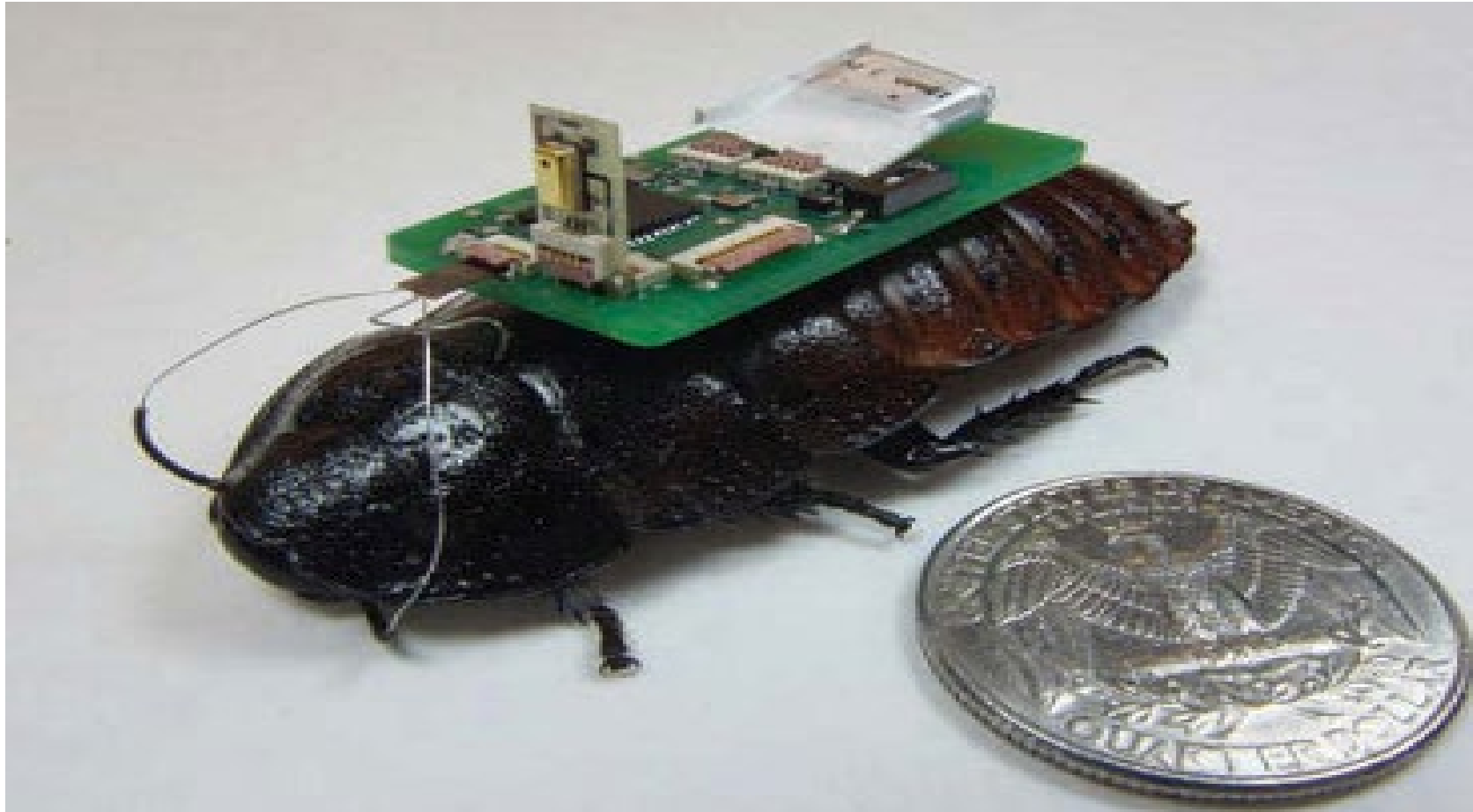
- An LED Digital Ceiling Light with Occupancy Sensor attached, providing control and power over the same infrastructure
- most modern LED ceiling fixtures support occupancy sensors
- modern lighting sensors integrate a variety of occupancy-sensing technologies, including Bluetooth low energy (BLE) and Wi-Fi



An LED Digital Ceiling Light with Occupancy Sensor

IoT Impact: Smart Creatures

- Connected Cows-Sparked, a Dutch company, developed a sensor that is placed in a cow's ear. The sensor monitors various health aspects of the cow as well as its location and transmits the data wirelessly for analysis by the farmer
- 200 MB data per year- needs infrastructure for data
- Researchers at North Carolina State University are working with Madagascar hissing cockroaches in the hopes of helping emergency personnel rescue survivors after a disaster.
- An electronic backpack attach with roach stimulate the cerci which makes cockroach to move forward.



IoT-Enabled Roach Can Assist in Finding Survivors After a Disaster
(Photo courtesy of Alper Bozkurt, NC State University)
<https://www.youtube.com/watch?v=gmbEX7zDzog>

Comparing OT and IT Network

Criterion	Industrial OT Network	Enterprise IT Network
Operational focus	Keep the business operating 24x7	Manage the computers, data, and employee communication system in a secure way
Priorities	1. Availability 2. Integrity 3. Security	1. Security 2. Integrity 3. Availability
Types of data	Monitoring, control, and supervisory data	Voice, video, transactional, and bulk data
Security	Controlled physical access to devices	Devices and users authenticated to the network
Implication of failure	OT network disruption directly impacts business	Can be business impacting, depending on industry, but workarounds may be possible
Network upgrades (software or hardware)	Only during operational maintenance windows	Often requires an outage window when workers are not onsite; impact can be mitigated
Security vulnerability	Low: OT networks are isolated and often use proprietary protocols	High: continual patching of hosts is required, and the network is connected to Internet and requires vigilant protection

Convergence of IT and OT

- More efficient and profitable business
- Lower costs through economy of scales
- Reduced inventory and improved delivery times
- With IT/OT convergence, IoT becomes fully supported by both groups.
- “best of both worlds” scenario, where solid industrial control systems reside on an open, integrated, and secure technology foundation

IoT Challenges

Challenges	Description
Scale	While the scale of IT networks can be large, the scale of OT can be several orders of magnitude larger. For example, one large electrical utility in Asia recently began deploying IPv6-based smart meters on its electrical grid. While this utility company has tens of thousands of employees (which can be considered IP nodes in the network), the number of meters in the service area is tens of millions. This means the scale of the network the utility is managing has increased by more than 1,000-fold!
Security	With more “things” becoming connected with other “things” and people, security is an increasingly complex issue for IoT. Your threat surface is now greatly expanded, and if a device gets hacked, its connectivity is a major concern. A compromised device can serve as a launching point to attack other devices and systems. IoT security is also pervasive across just about every facet of IoT
Privacy	As sensors become more prolific in our everyday lives, much of the data they gather will be specific to individuals and their activities. This data can range from health information to shopping patterns and transactions at a retail establishment. For businesses, this data has monetary value. Organizations are now discussing who owns this data and how individuals can control whether it is shared and with whom

IoT Challenges

Challenges	Description
Big data and data analytics	IoT and its large number of sensors is going to trigger a deluge of data that must be handled. This data will provide critical information and insights if it can be processed in an efficient manner. The challenge, however, is evaluating massive amounts of data arriving from different sources in various forms and doing so in a timely manner
Interoperability	As with any other nascent technology, various protocols and architectures are jockeying for market share and standardization within IoT. Some of these protocols and architectures are based on proprietary elements, and others are open. Recent IoT standards are helping minimize this problem, but there are often various protocols and implementations available for IoT networks. The prominent protocols and architectures—especially open, standards-based implementations

Summary

- IoT is about connecting the unconnected, enabling smart objects to communicate with other objects, systems, and people
- High-level use cases discussed to show the impact of IoT
- Difference between IoT and Digitization was discussed
- Challenges faced by IoT

- Active mode (AM)
 - All clocks are active
- Low-power mode 0 (LPM0)
 - CPU is disabled
 - ACLK and SMCLK remain active, MCLK is disabled
 - FLL loop control remains active
- Low-power mode 1 (LPM1)
 - CPU is disabled
 - FLL loop control is disabled
 - ACLK and SMCLK remain active, MCLK is disabled
- Low-power mode 2 (LPM2)
 - CPU is disabled
 - MCLK and FLL loop control and DCOCLK are disabled
 - DCO's dc-generator remains enabled
 - ACLK remains active
- Low-power mode 3 (LPM3)
 - CPU is disabled
 - MCLK, FLL loop control, and DCOCLK are disabled
 - DCO's dc generator is disabled
 - ACLK remains active
- Low-power mode 4 (LPM4)
 - CPU is disabled
 - ACLK is disabled
 - MCLK, FLL loop control, and DCOCLK are disabled
 - DCO's dc generator is disabled
 - Crystal oscillator is stopped
 - – Complete data retention